Networks Lab Assignment 2 Report

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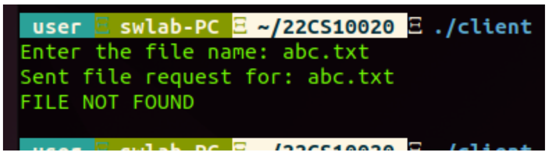
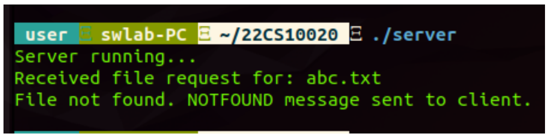
1. Client and Server Terminals

Scenario 1: File Not Available in the Local Directory

Below are the terminal screenshots for the client and server when the file was not available in the local directory:



(a) Client terminal output.

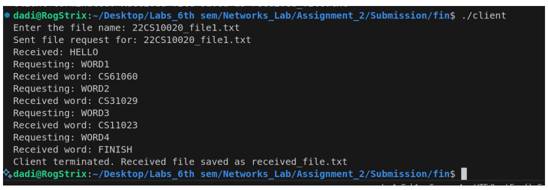


(b) Server terminal output.

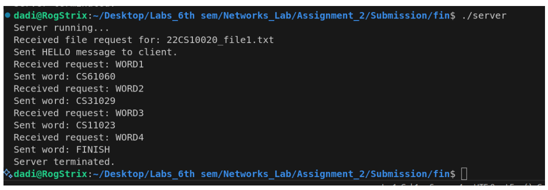
Figure 1: Client and server terminals when the file was not available.

Scenario 2: File Available in the Local Directory

Below are the terminal screenshots for the client and server when the file was available in the local directory:



(a) Client terminal output.

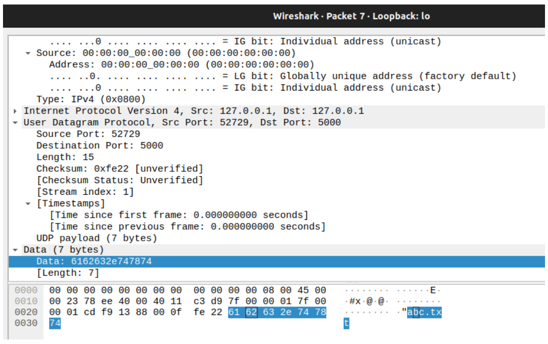


(b) Server terminal output.

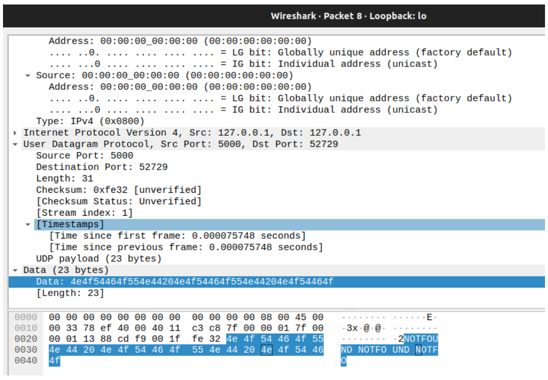
Figure 2: Client and server terminals when the file was available.

2. Packet Exchange Screenshots

Scenario 1: File Not Available in the Local Directory



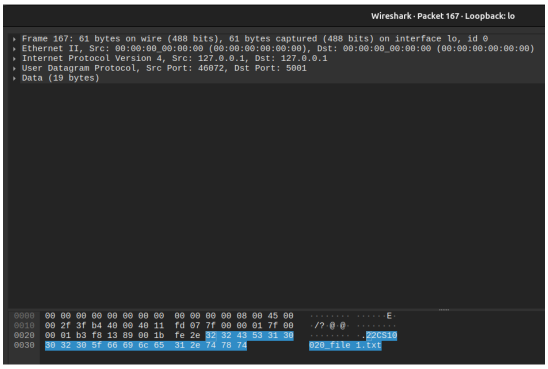
(a) First packet(file name).



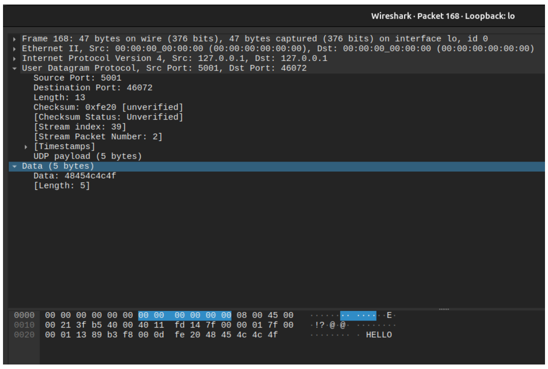
(b) Second packet(not found message).

Figure 3: Packets exchanged when the file was not available.

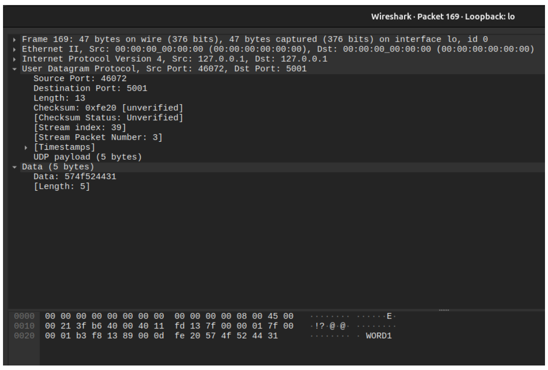
Scenario 2: File Available in the Local Directory



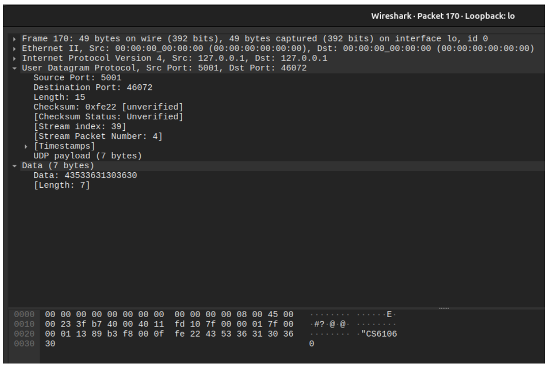
(a) Packet 1: Client sends filename request "22CS10020_file1.txt"



(b) Packet 2: Server responds with "HELLO"

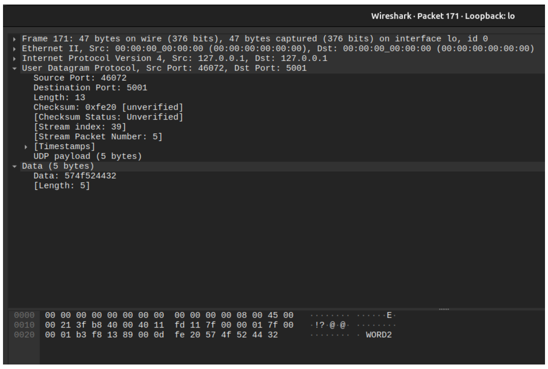


(c) Packet 3: Client requests WORD1

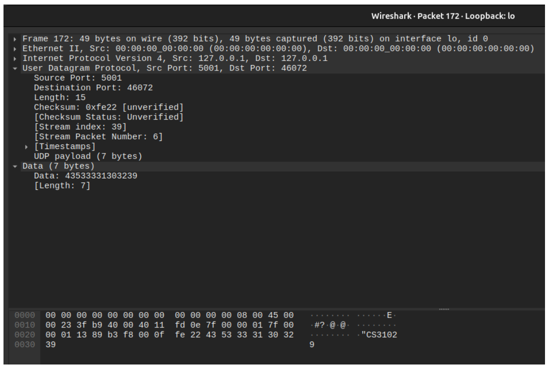


(d) Packet 4: Server sends first word "CS61060"

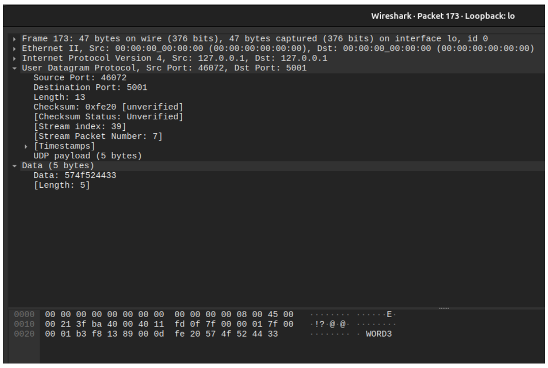
Figure 4: Initial phase: File request and first word transfer



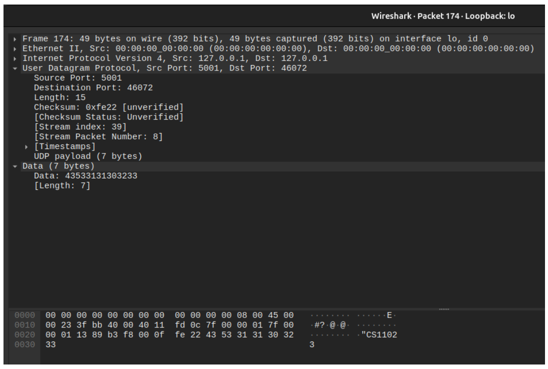
(a) Packet 5: Client requests WORD2



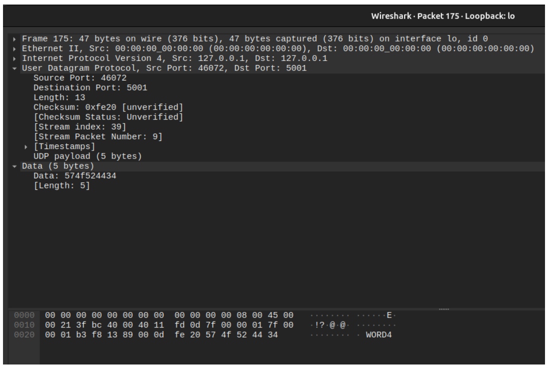
(b) Packet 6: Server sends second word "CS31029"



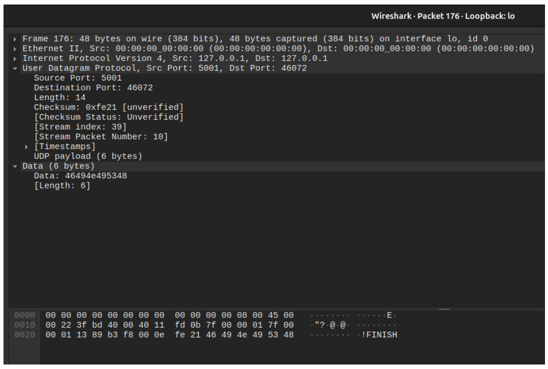
(c) Packet 7: Client requests WORD3



(d) Packet 8: Server sends third word "CS11023"



(a) Packet 9: Client requests WORD4



(b) Packet 10: Server sends "FINISH", ending transfer

Figure 6: Final phase: Last word request and completion

3. Protocol Used

Task: Identify the protocol used for communication.
Protocol: *UDP*

4. Source and Destination Details

Task: Identify the IP addresses and ports of the source and destination.

When the File Was Available	
Client IP and Port:	*127.0.0.1, PORT: 46072*
Server IP and Port:	*127.0.0.1, PORT: 5001*
When the File Was Not Available	
Client IP and Port:	*127.0.0.1, PORT: 52729*
Server IP and Port:	*127.0.0.1, PORT: 5001*

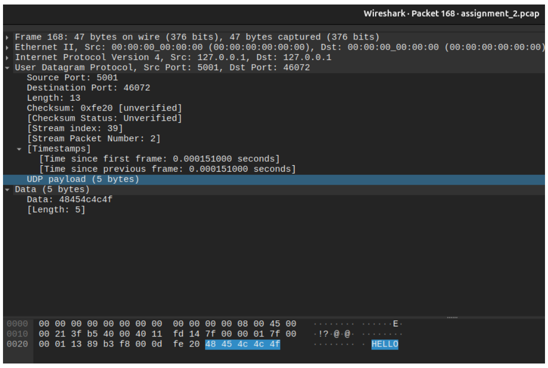
Table 1: Client and Server IP and Port Information

5. Packet Sizes

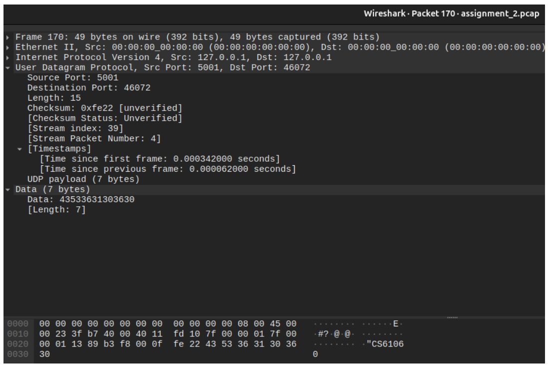
Task 1: Measure the size of the FILENAME request sent by the client.
Size: *19 bytes*
Task 2: Measure the size of the server's responses for HELLO and the first word ($WORD1$).
HELLO Response Size: *47 bytes on the wire (376 bits), 47 bytes (376 bits) captured on interface `lo`, id 0*
$WORD1$ **Response Size:** *49 bytes on the wire (376 bits), 47 bytes (376 bits) captured on interface `lo`, id 0*

6. Payload Inspection

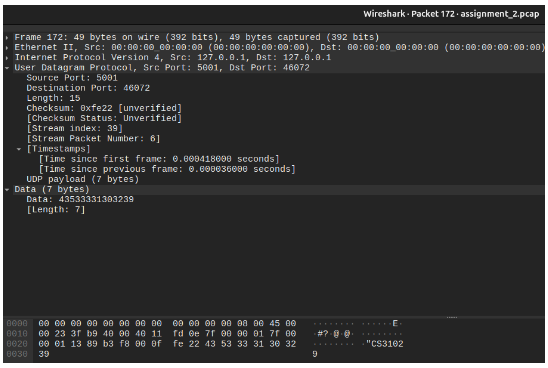
Task: Inspect the payload of packets where words are transmitted.
Payload Details: The UDP payloads of packets transmitting the words are shown below. Each packet contains a specific payload, as visible in the captured data. The hexadecimal representation and its ASCII equivalent are included for each word.



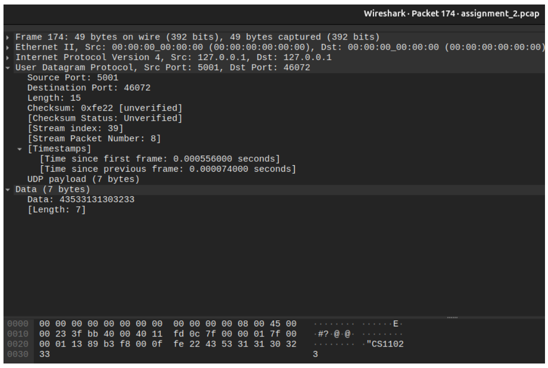
(a) UDP payload containing the word `HELLO`.



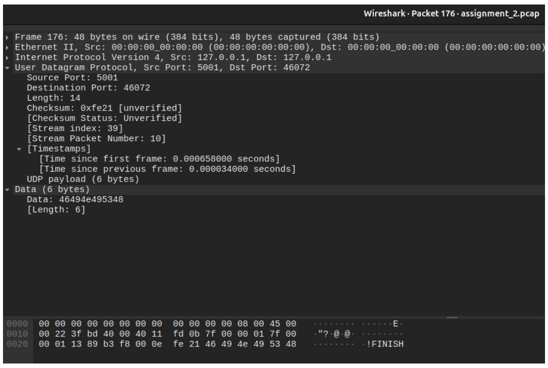
(b) UDP payload containing the word `CS61060`.



(a) UDP payload containing the word `CS31029`.



(b) UDP payload containing the word `CS11023`.



(a) UDP payload containing the word `FINISH`.

7. Total File Transfer Time

Task: Measure the total time taken for the file transfer from start to finish.
Total Time: *0.000507 seconds*

8. Average Packet Size

Task: Calculate the average size of each packet during communication.
Average Packet Size: *47.78 bytes*